Vietnam National University of HCMC

International University

School of Computer Science and Engineering

Program of MIT

Course: Advanced Computer Graphics (IT516)

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Transformation and Perspective (24/3/2026)

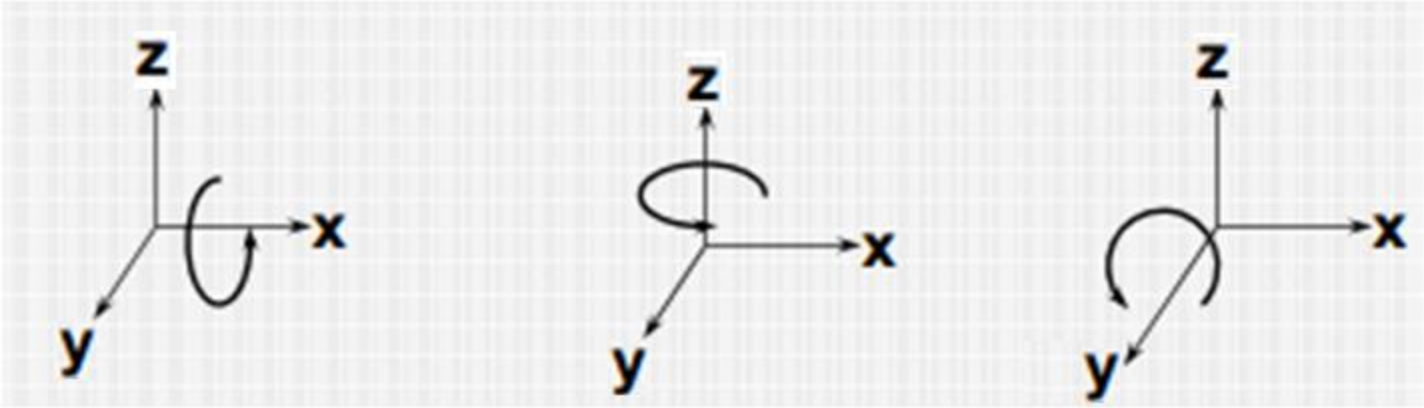
- Textbook: chapter 7
- Mathematics for CG: chapter 9 (9.6)
- Intro to CG using Java 2D & 3D: chapter 5

Arbitrary *3D Rotation*

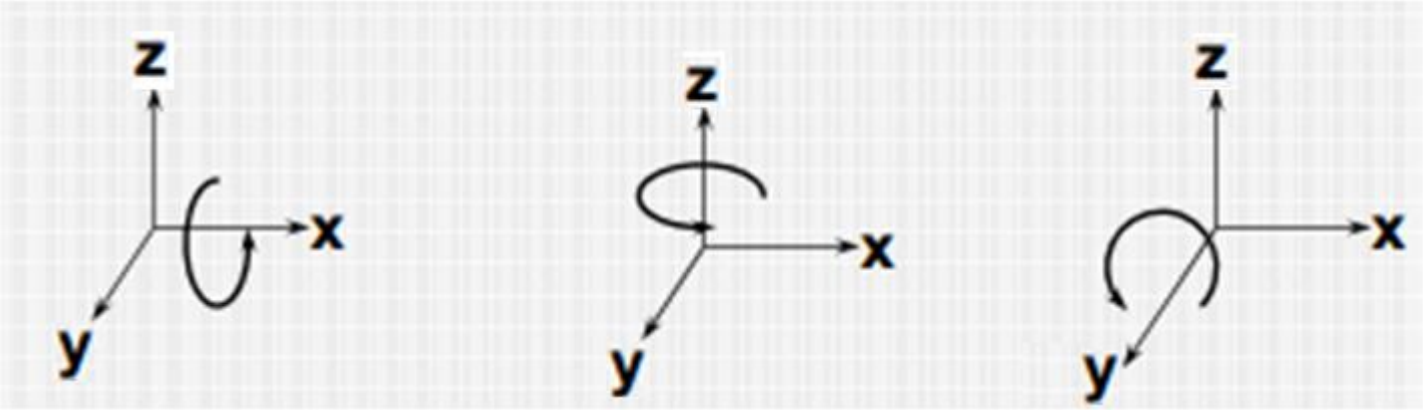
- What is its inverse?
- What is its transpose?
- Can we constructively elucidate this relationship?

Recall: *3D Rotation*

- In 2D, the only rotation possible was about the origin.
- In 3D, there are 3 possible rotations, one about each of the x, y and z axes.
- Positive rotations are anti-clockwise, negative rotations are clockwise, when looking down a positive axis towards the origin



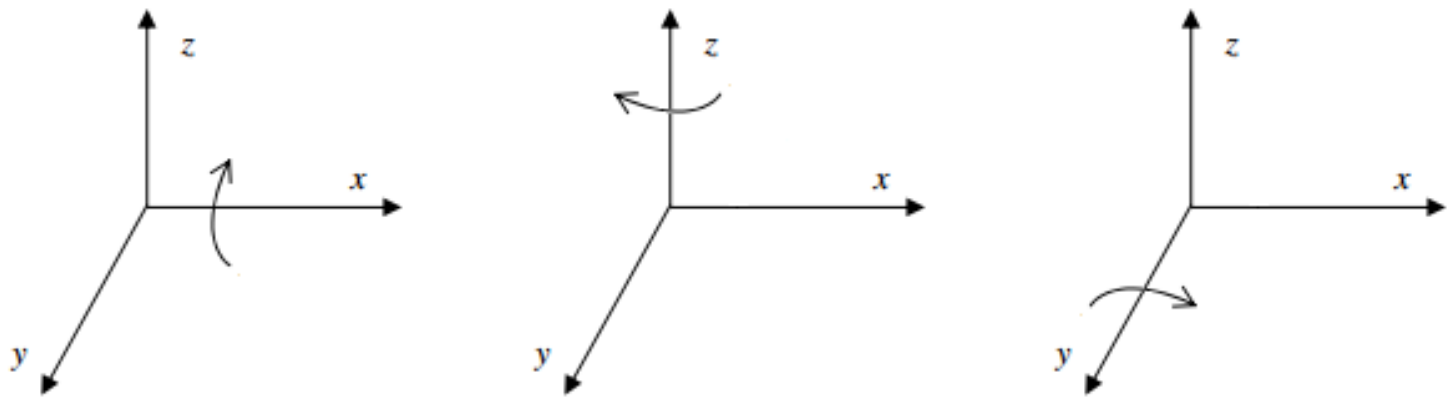
Recall: *3D Rotation*



Anti-clockwise (+θ)

- Rotate x-axis: $x' = x$; $y' = y\cos\theta - z\sin\theta$; $z' = y\sin\theta + z\cos\theta$
- Rotate y-axis: $y' = y$; $x' = x\cos\theta + z\sin\theta$; $z' = -x\sin\theta + z\cos\theta$
- Rotate z-axis: $z' = z$; $x' = x\cos\theta - y\sin\theta$; $y' = x\sin\theta + y\cos\theta$

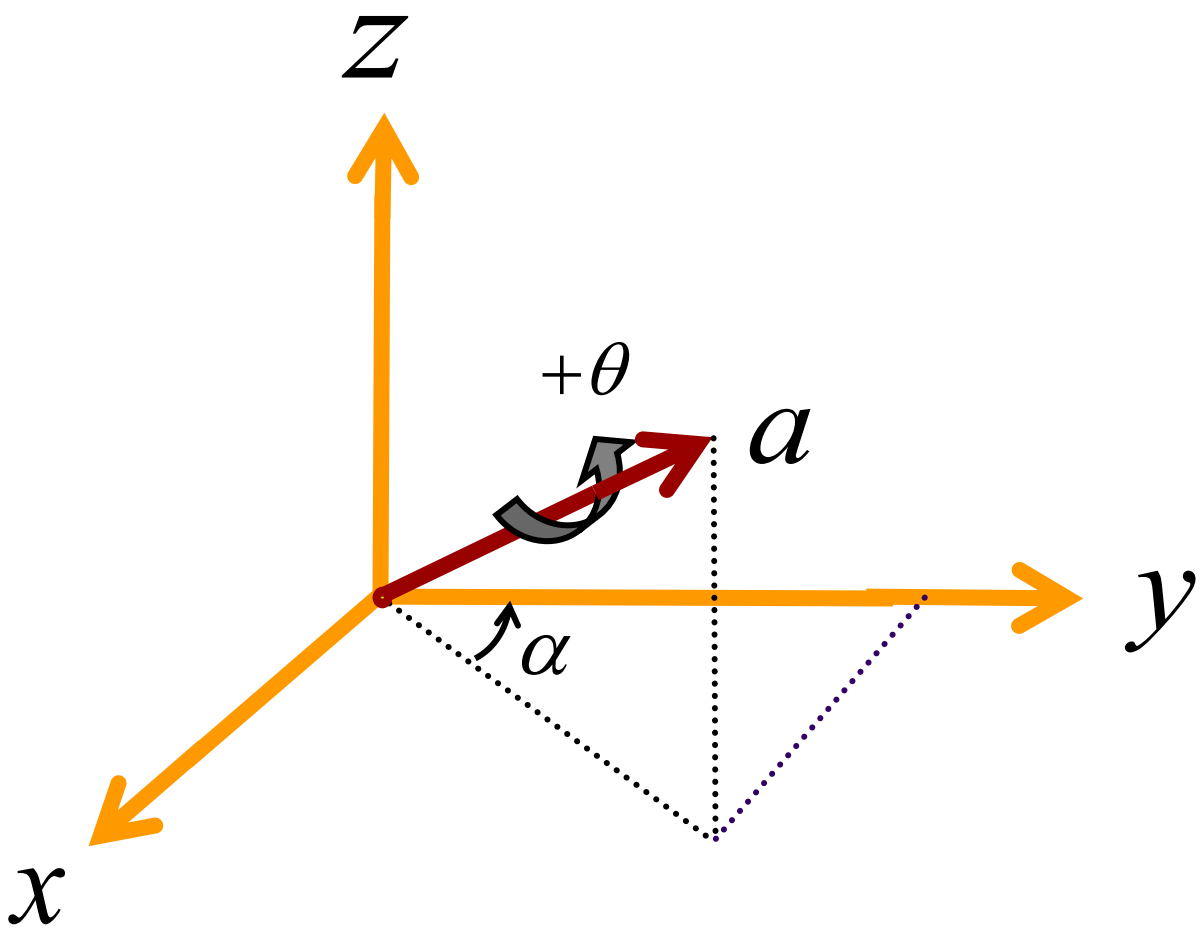
Recall: *3D Rotation*



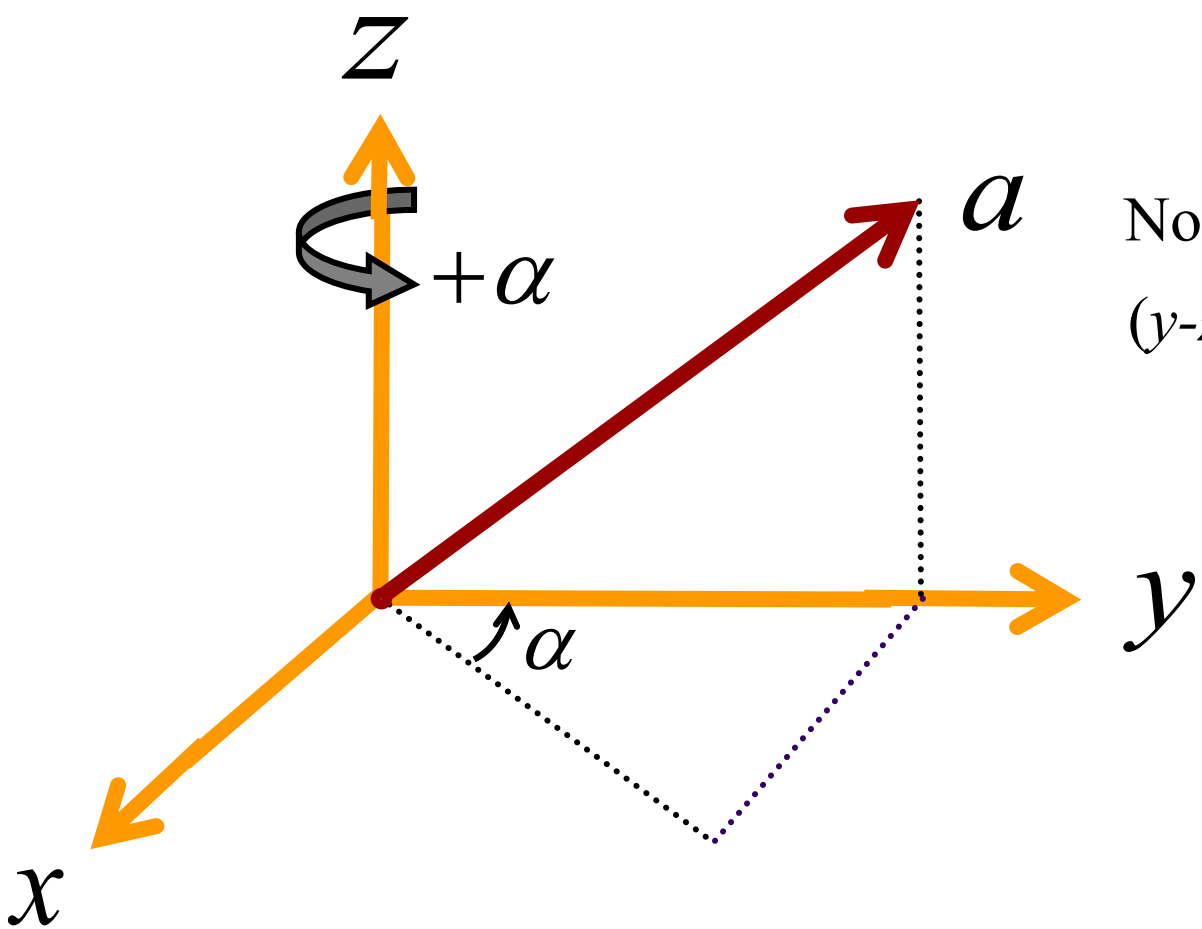
Clockwise (-θ)

- Rotate x-axis: $x' = x$; $y' = y\cos\theta - z\cos\theta$; $z' = y\sin\theta + z\cos\theta$
- Rotate y-axis: $y' = y$; $x' = z\sin\theta + x\cos\theta$; $z' = z\cos\theta + x\cos\theta$
- Rotate z-axis: $z' = z$; $x' = x\cos\theta - y\sin\theta$; $y' = x\sin\theta + y\cos\theta$

Rotate $+\theta$ *about axis* a: $R_a(+\theta)$

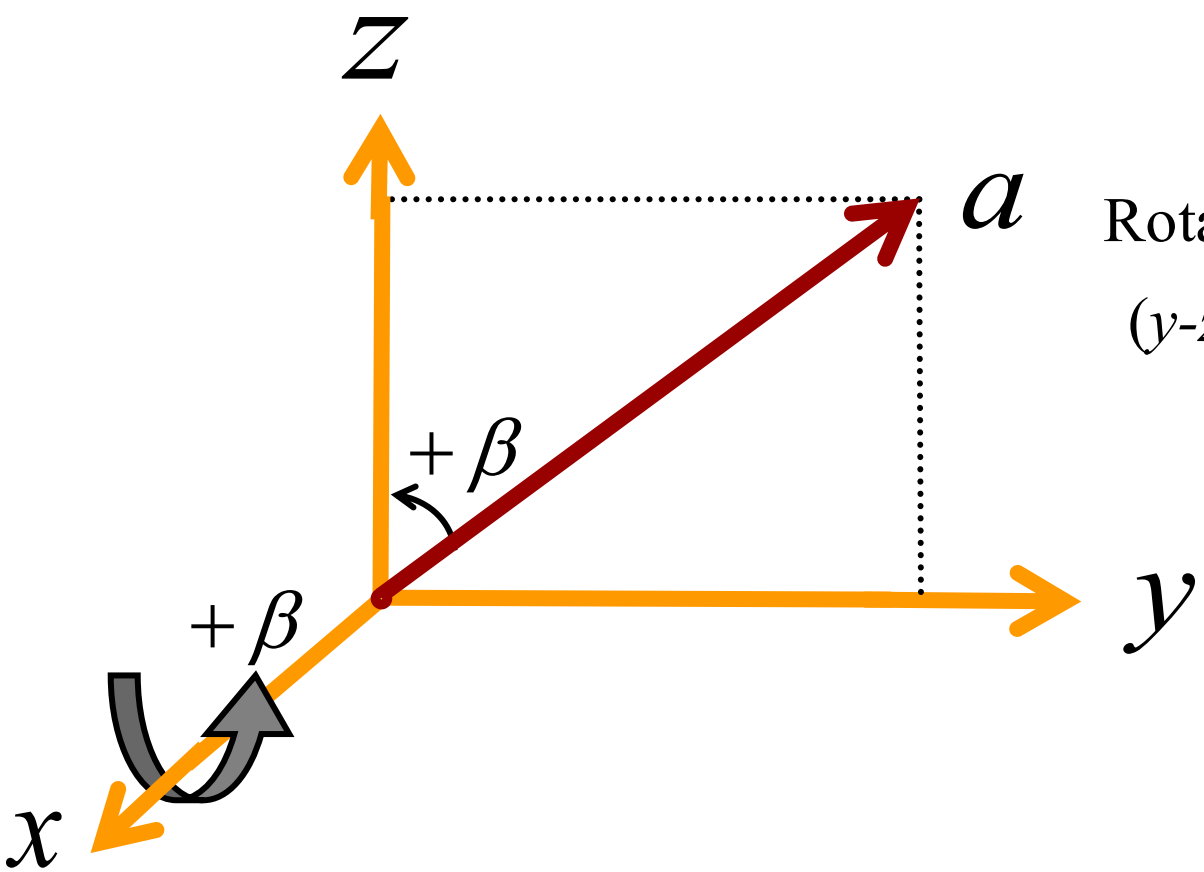


First, *Rotate about z by* $+\alpha$: $R_z(+\alpha)$



Now in the (y-z)-plane

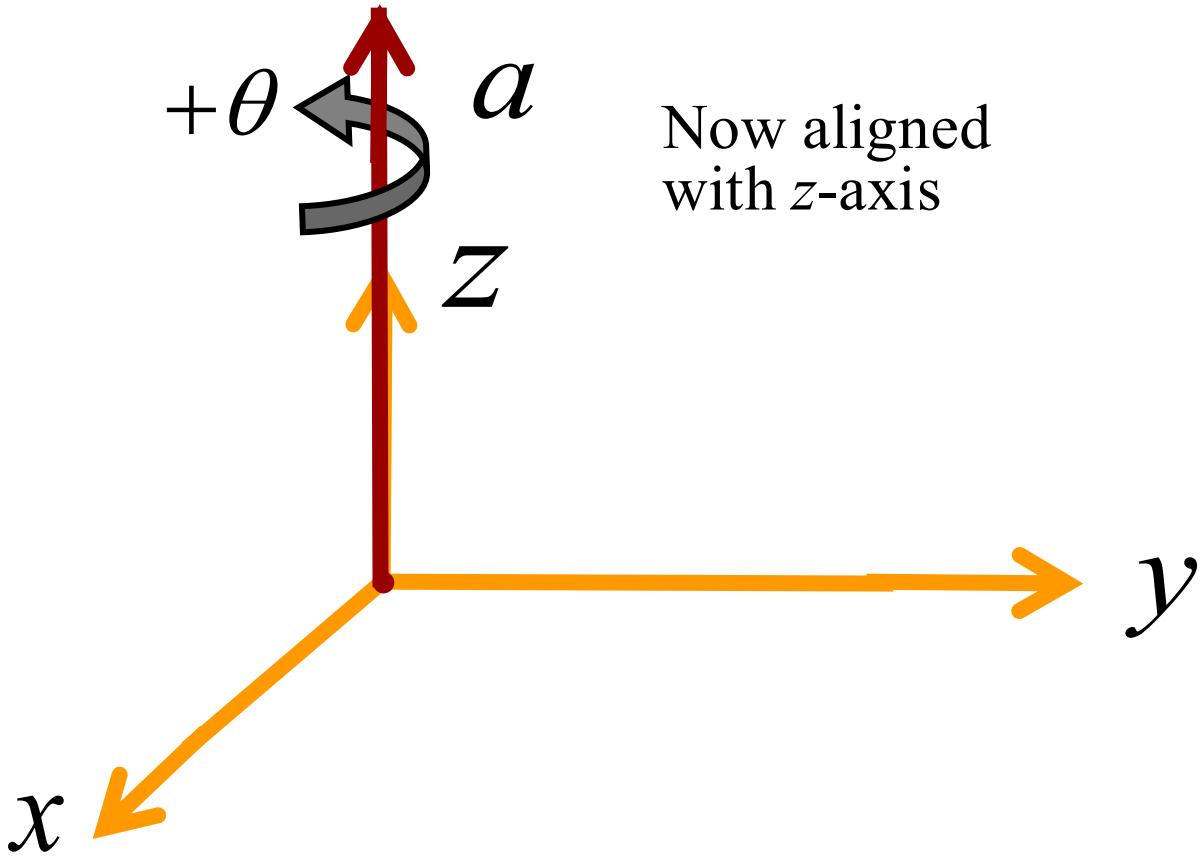
Then *Rotate about x by* $+\beta$: $R_x(+\beta)$



Rotate in the

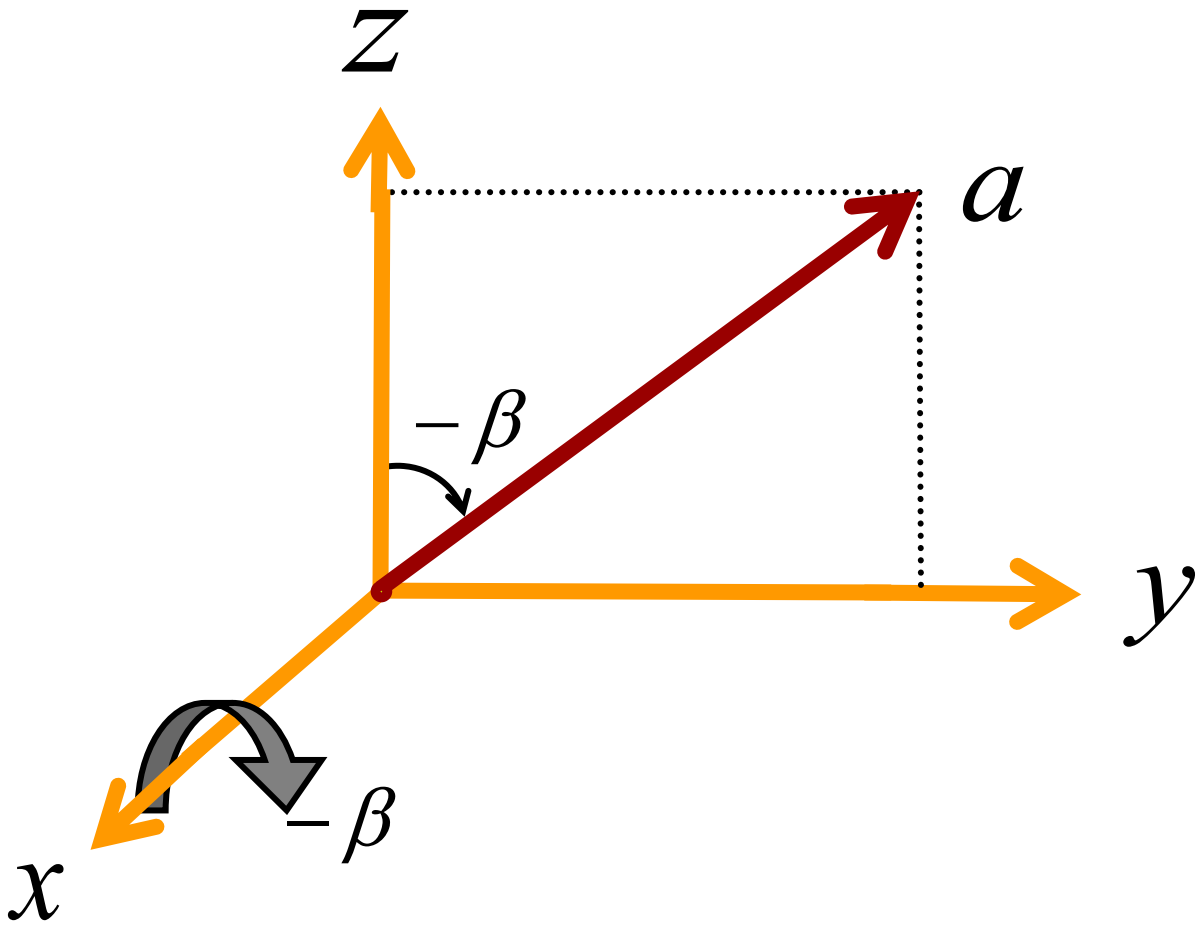
(y-z)-plane

Now, $+\theta$ *Rotation about z-axis*: $R_z(+\theta)$



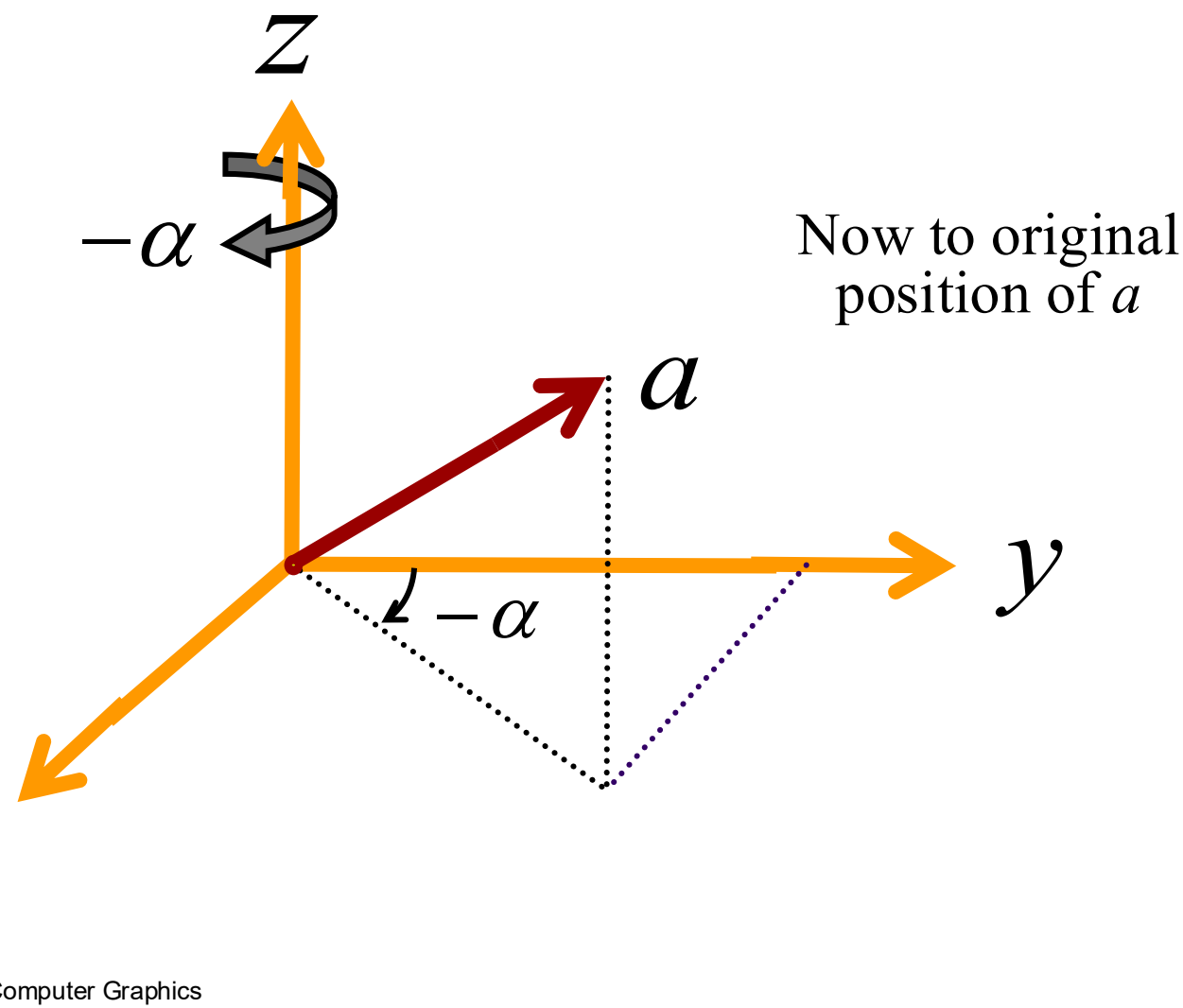
Now aligned with z-axis

Then *rotate* about x by $-\beta$: $R_x(-\beta)$

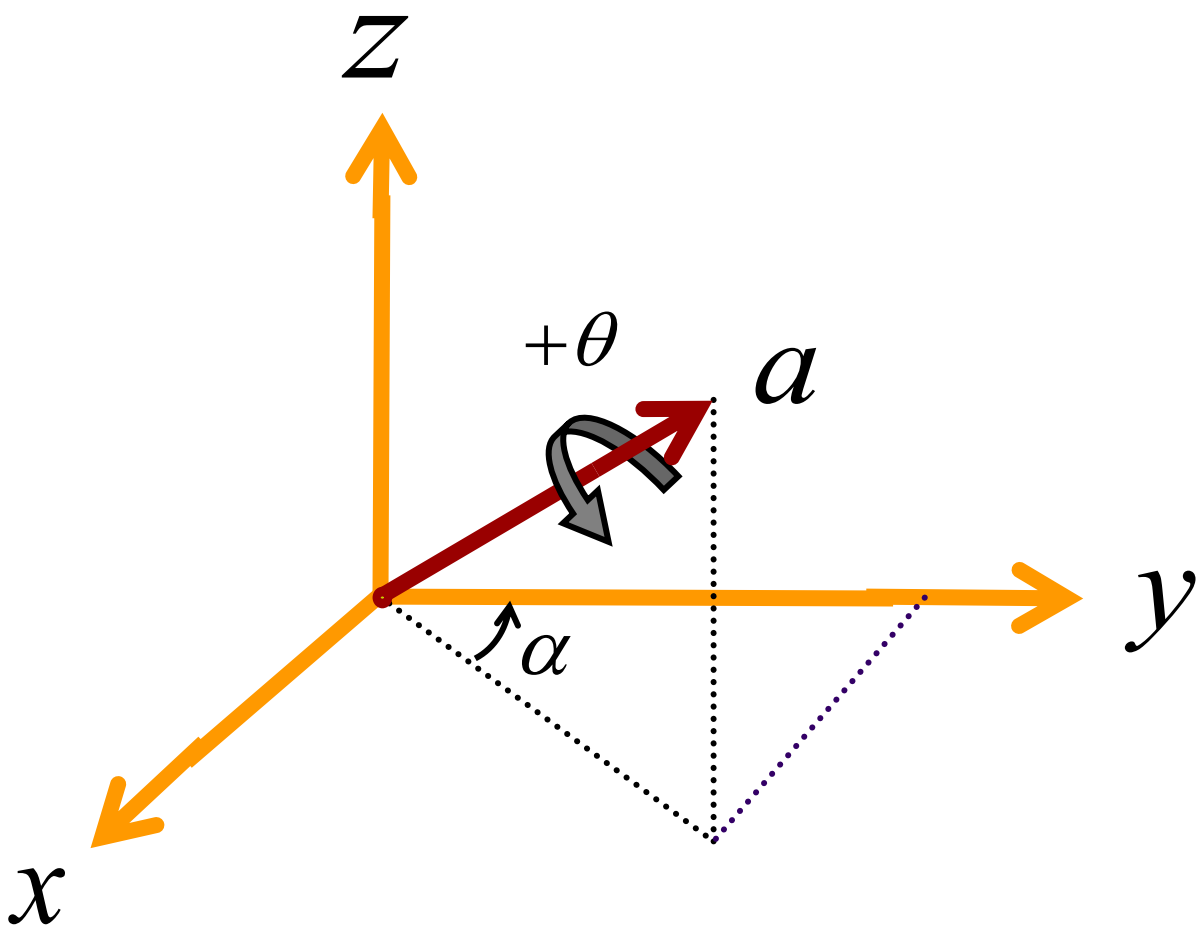


Rotate again in the (y-z)-plane

Now, $+\theta$ *Rotation* about z by $-\alpha$: $R_z(-\alpha)$



We Effected $+\theta$ *rotation* about Arbitrary *axis* a: $R_a(+\theta)$



We Effected $+\theta$ *rotation* about Arbitrary *axis a*:

$$R_a(+\theta)$$

$$R_a(+\theta) = R_z(-\alpha)\ R_x(-\beta)\ R_z(+\theta)$$

$$\times\ R_x(+\beta)\ R_z(+\alpha)$$

Rotation about Arbitrary Axis

- Rotation about *a-axis* effected by (nonunique) composition of 5 elementary *rotations*
- We show arbitrary *rotation* as succession of 5 *rotations* about *principal axes*

In matrix terms, $R_a(+\theta) =$

$$R_a(+\theta) = \overbrace{\begin{bmatrix} \cos(-\alpha) & -\sin(-\alpha) & 0 & 0 \\ \sin(-\alpha) & \cos(-\alpha) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}^{R_Z(-\alpha)} \overbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(-\beta) & -\sin(-\beta) & 0 \\ 0 & \sin(-\beta) & \cos(-\beta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}^{R_X(-\beta)}$$

$$\times \underbrace{\begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{R_Z(+\theta)} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\beta & -\sin\beta & 0 \\ 0 & \sin\beta & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{R_X(+\beta)} \underbrace{\begin{bmatrix} \cos\alpha & -\sin\alpha & 0 & 0 \\ \sin\alpha & \cos\alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{R_Z(+\alpha)}$$

Similarly, $R_a^{-1}(+\theta) = R_a^{-1}(-\theta)$, so

$$R_a(-\theta) = \begin{bmatrix} \cos(-\alpha) & -\sin(-\alpha) & 0 & 0 \\ \sin(-\alpha) & \cos(-\alpha) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(-\beta) & -\sin(-\beta) & 0 \\ 0 & \sin(-\beta) & \cos(-\beta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\times \underbrace{\begin{bmatrix} \cos\theta & \sin\theta & 0 & 0 \\ -\sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{R_z(-\theta)} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\beta & -\sin\beta & 0 \\ 0 & \sin\beta & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 & 0 \\ \sin\alpha & \cos\alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Recall, $[AB]^{\mathbf{t}} = B^{\mathbf{t}} A^{\mathbf{t}}$

Consequently, for $A = R^t \underline{M} R$,

$A^t = R^t \underline{M^t} R$ because,

$$\left[R^t (M R)\right]^t = \left[M \cdot R\right]^t \left[R^t\right]^t$$

$$= R^t \; M^t \; R$$

It follows directly that,

$$\left[R^t S^t \underline{M} S R\right]^t = R^t S^t M^t \underline{S} R$$

$$\therefore \quad R_a^{-1}(+\theta) = R_a^t(\theta)$$

$$R_a(-\theta) = \begin{bmatrix} \cos(-\alpha) & -\sin(-\alpha) & 0 & 0 \\ \sin(-\alpha) & \cos(-\alpha) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos(-\beta) & -\sin(-\beta) & 0 \\ 0 & \sin(-\beta) & \cos(-\beta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\times \underbrace{\begin{bmatrix} \cos\theta & \sin\theta & 0 & 0 \\ -\sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{R_Z^t(\theta)} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\beta & -\sin\beta & 0 \\ 0 & \sin\beta & \cos\beta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 & 0 \\ \sin\alpha & \cos\alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Constructively, we have shown,

$$R_a^{-1}(\theta) = R_a^t(\theta)$$

This will be useful later

3D *Translation in x*

$$T_x(d_x) = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x+d_x \\ y \\ z \\ 1 \end{bmatrix}$$

3D *Translation in y*

$$T_y(d_y) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y+d_y \\ z \\ 1 \end{bmatrix}$$

3D *Translation in z*

$$T_z(d_z) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z+d_z \\ 1 \end{bmatrix}$$

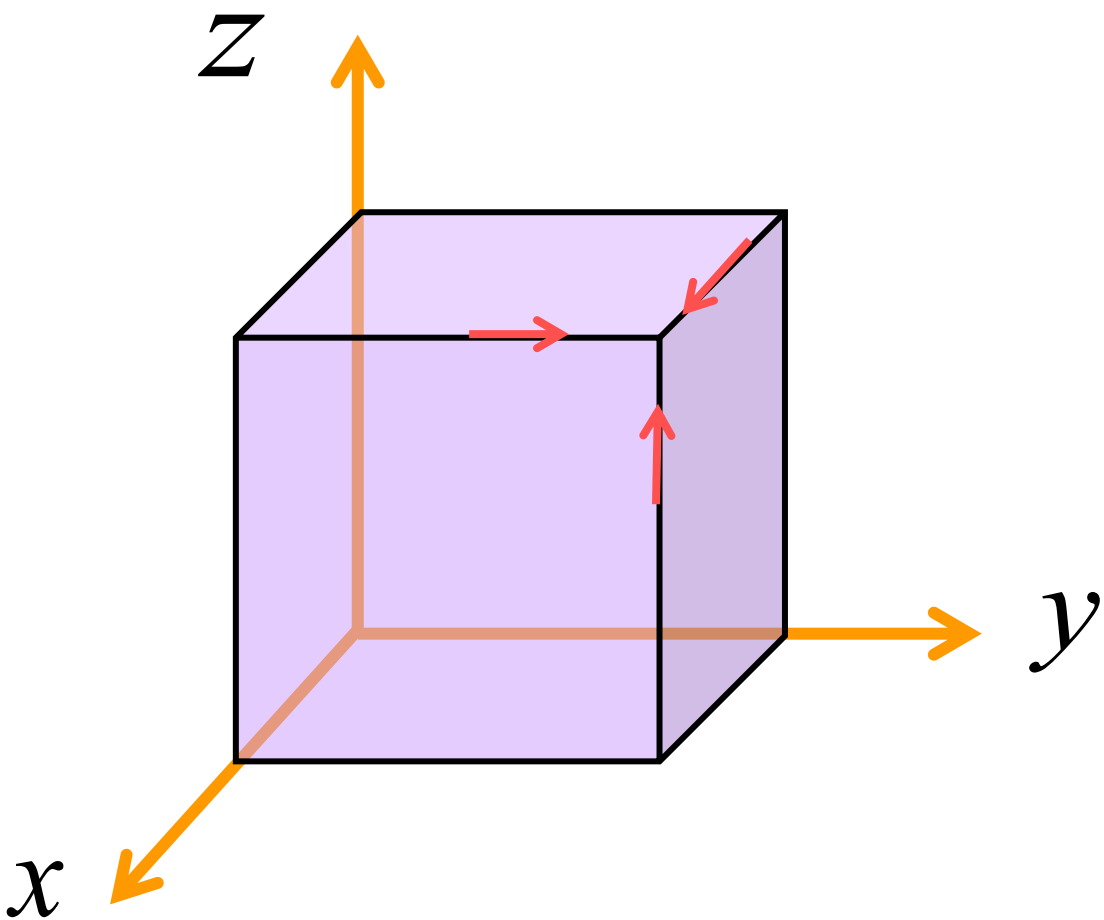
3D *Shear in x -direction*

$$Sh_x(a) = \begin{bmatrix} 1 & a & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x+ay \\ y \\ z \\ 1 \end{bmatrix}$$

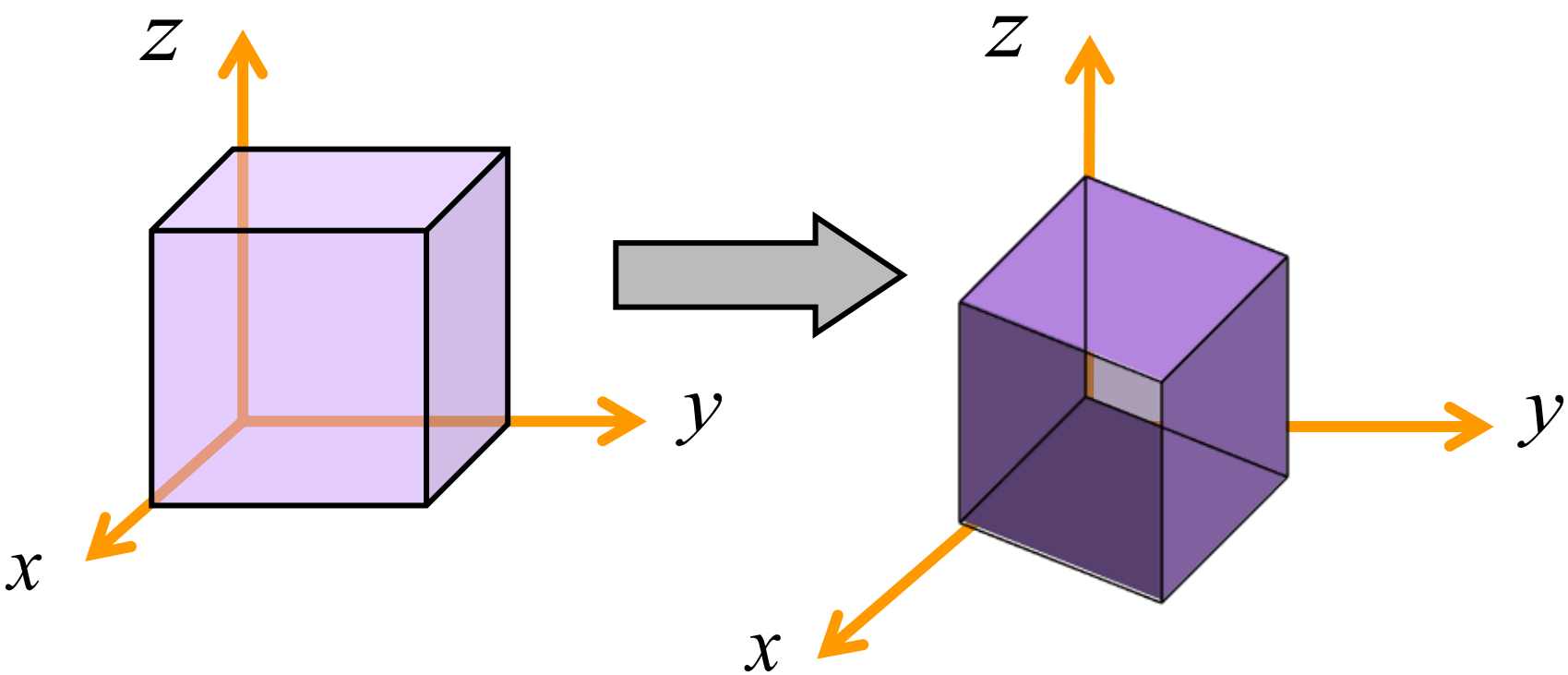
3D *Shear in x -direction*

$$Sh_x(b) = \begin{bmatrix} 1 & 0 & b & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x+bz \\ y \\ z \\ 1 \end{bmatrix}$$

3D *Shears*: Clamp a *Principal Plane*, shear in other to *DoF (degree of freedom)*



$$Sh_x(a) = \begin{bmatrix} 1 & a & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x+ay \\ y \\ z \\ 1 \end{bmatrix}$$



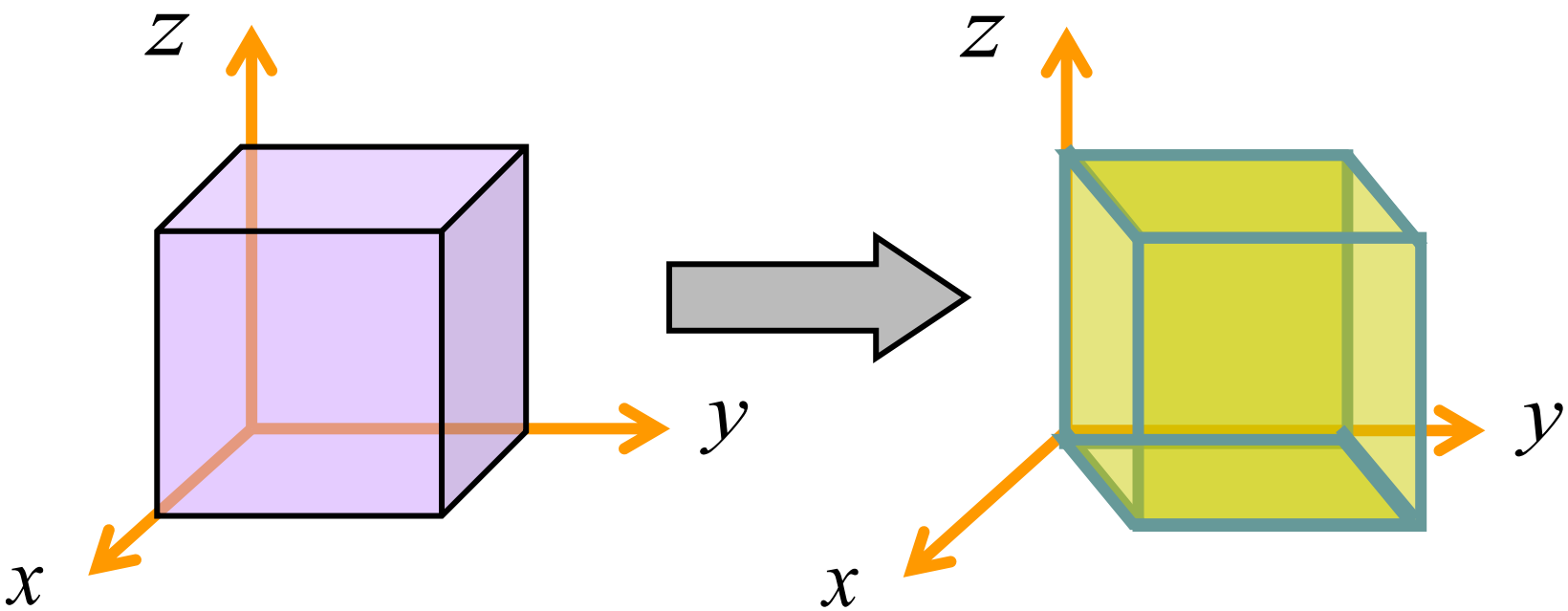
3D *Shear in y -direction*

$$Sh_y(c) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & c & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y+cz \\ z \\ 1 \end{bmatrix}$$

3D *Shear in y -direction*

$$Sh_y(d) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ d & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ dx+y \\ z \\ 1 \end{bmatrix}$$

$$Sh_y(d) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ d & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ dx+y \\ z \\ 1 \end{bmatrix}$$



3D *Shear in z-direction*

$$Sh_z(e) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ e & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ ex+z \\ 1 \end{bmatrix}$$

3D *Shear in z*

$$Sh_z(e) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ e & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ ex+z \\ 1 \end{bmatrix}$$

3D *Shear in z*

$$Sh_z(f) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & f & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ fy+z \\ 1 \end{bmatrix}$$

What About Elementary *Inverses*?

- *Scale*
- *Shear*
- *Rotation*
- *Translation*

What About Elementary *Inverses*?

- Normally, we always invert a matrix algebraically. Now we can use geometry if we know what the transform does. E.g:
  - The inverse of scale (s_x, s_y, s_z)is scale($1/s_x$,$1/s_y$,$1/s_z$).
  - The inverse of a rotation is the same rotation with the opposite sign on the angle.
  - The inverse of a translation is a translation in the opposite direction.
  - …
  - If we have a series of matrices $M = M_1 M_2 \cdots M_n$ then $[M]^{-1} = [M_n]^{-1} \ldots [M_2]^{-1} [M_1]^{-1}$

Scale Inverse

$$\begin{bmatrix} \lambda & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\lambda & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \lambda & 0 \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1/\lambda & 0 \\ 0 & 1 \end{bmatrix}$$

Shear Inverse

$$\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -a \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -b & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Shear Inverse

$$\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -a \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 \\ -b & 1 \end{bmatrix}$$

Rotation Inverse

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

Rotation Inverse

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} =$$

$$= \begin{bmatrix} (\cos^2\theta + \sin^2\theta) & (\cos\theta\sin\theta - \cos\theta\sin\theta) \\ (\cos\theta\sin\theta - \cos\theta\sin\theta) & (\sin^2\theta + \cos^2\theta) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Rotation Inverse

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}^{-1} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$$

Translation Inverse

$$\begin{bmatrix} 1 & 0 & d_x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -d_x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 1 & 0 & (-d_x + d_x) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Translation Inverse

$$\begin{bmatrix} 1 & 0 & d_x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & -d_x \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Double Shear in x and y

$$\overset{Sh_y(b)}{\begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix}} \overset{Sh_x(a)}{\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}} = \begin{bmatrix} 1 & a \\ b & (1+ab) \end{bmatrix}$$

$$\underset{Sh_x(a)}{\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}} \underset{Sh_y(b)}{\begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix}} = \begin{bmatrix} (1+ab) & a \\ b & 1 \end{bmatrix}$$

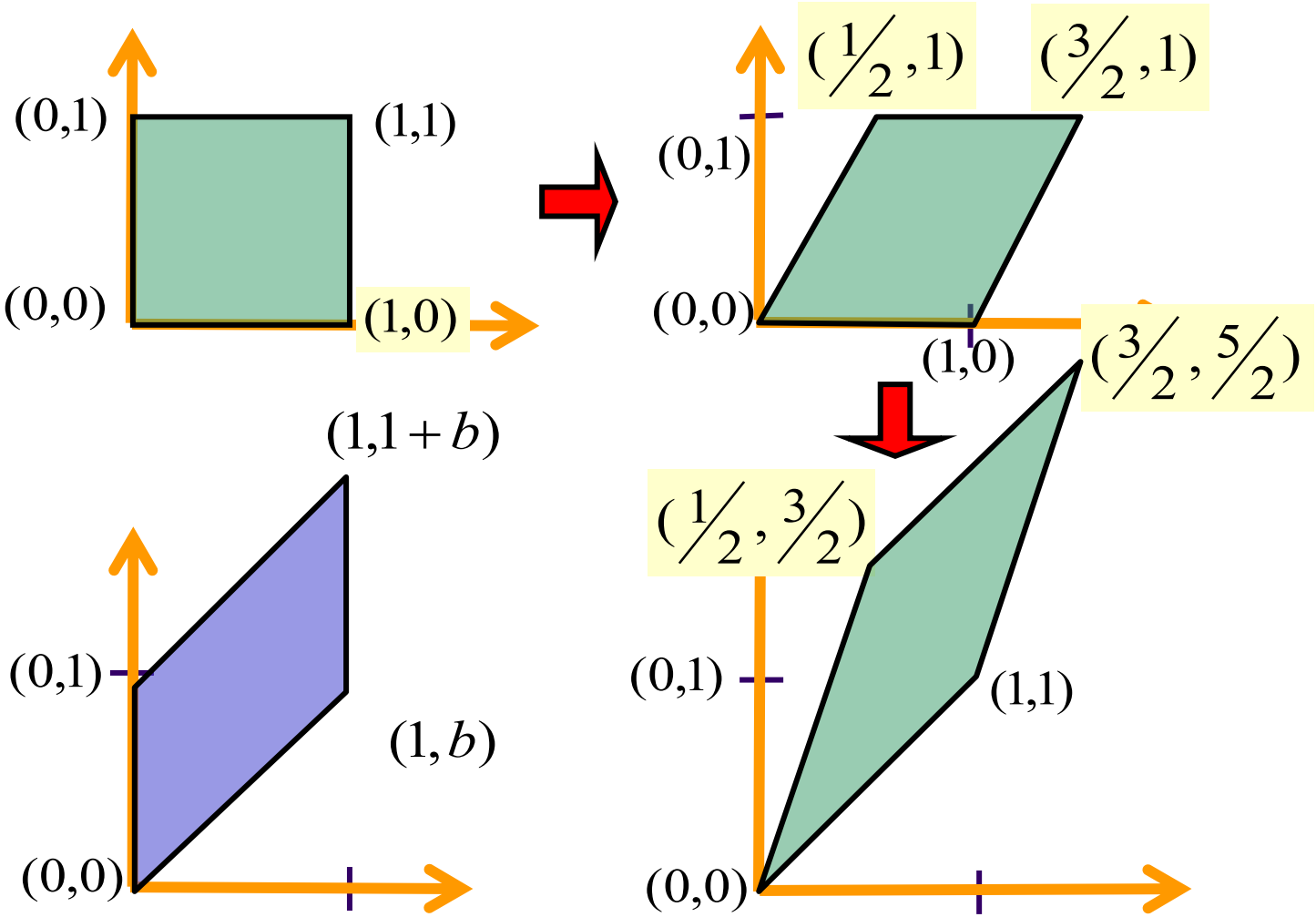
Ex: a=0.5 and b=1.0

$$\overset{Sh_y(b)}{\begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix}} \overset{Sh_x(a)}{\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}} = \begin{bmatrix} 1 & a \\ b & (1+ab) \end{bmatrix}$$

$$\underset{Sh_x(a)}{\begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}} \underset{Sh_y(b)}{\begin{bmatrix} 1 & 0 \\ b & 1 \end{bmatrix}} = \begin{bmatrix} (1+ab) & a \\ b & 1 \end{bmatrix}$$

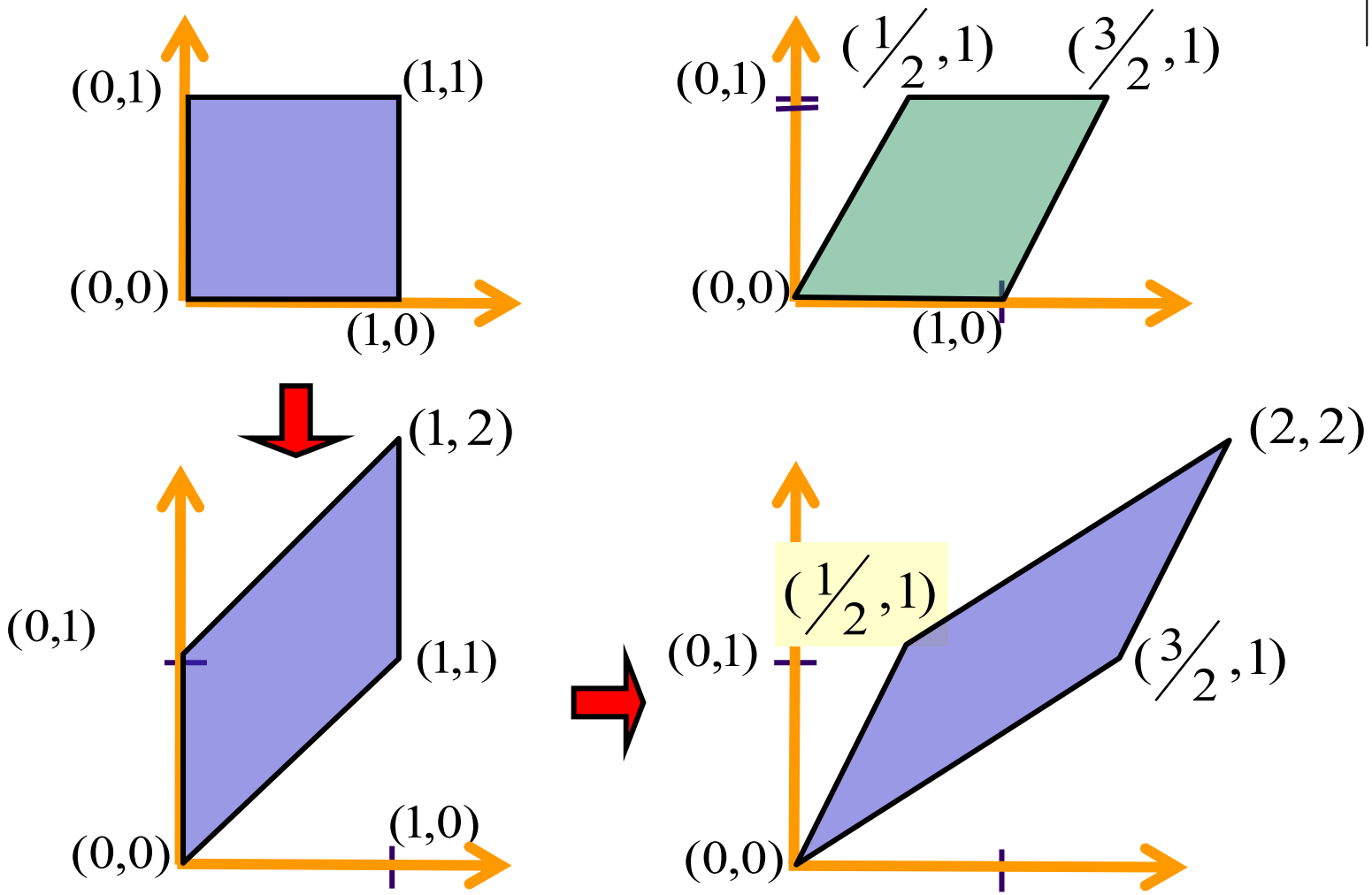
Shear in x then, y:

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1/2 \\ 1 & 3/2 \end{bmatrix}$$



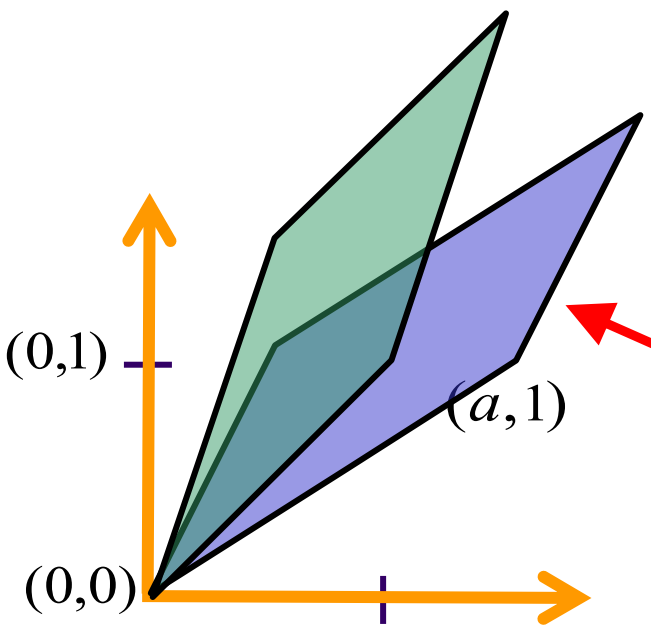
Shear in x then, y :

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1/2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1/2 \\ 1 & 3/2 \end{bmatrix}$$



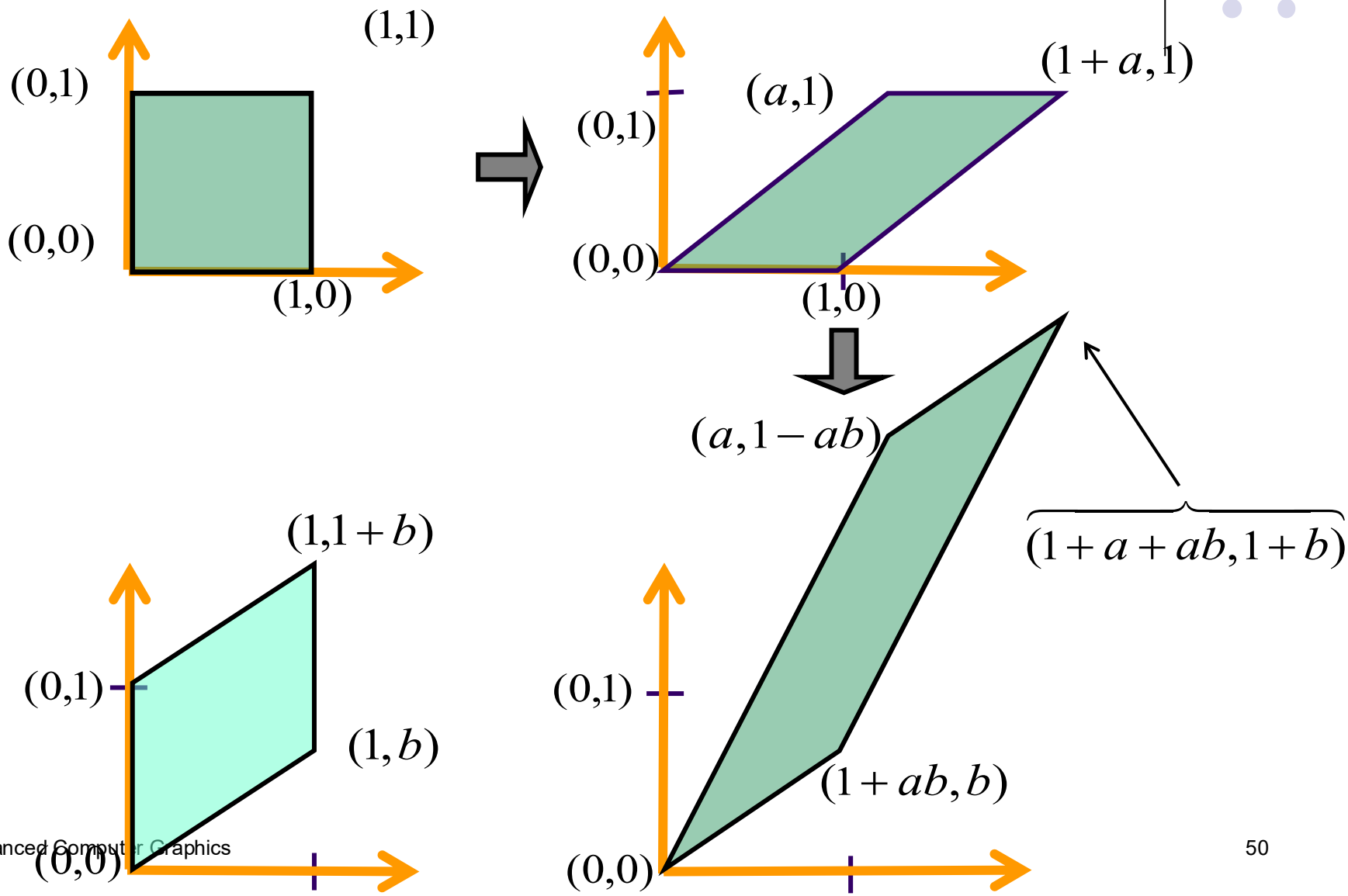
Shears in x and y do not Commute

$$Sh\ in\ x_{(a=1/2)}\ ,\ \text{then}\ Sh\ y_{(b=1)}$$

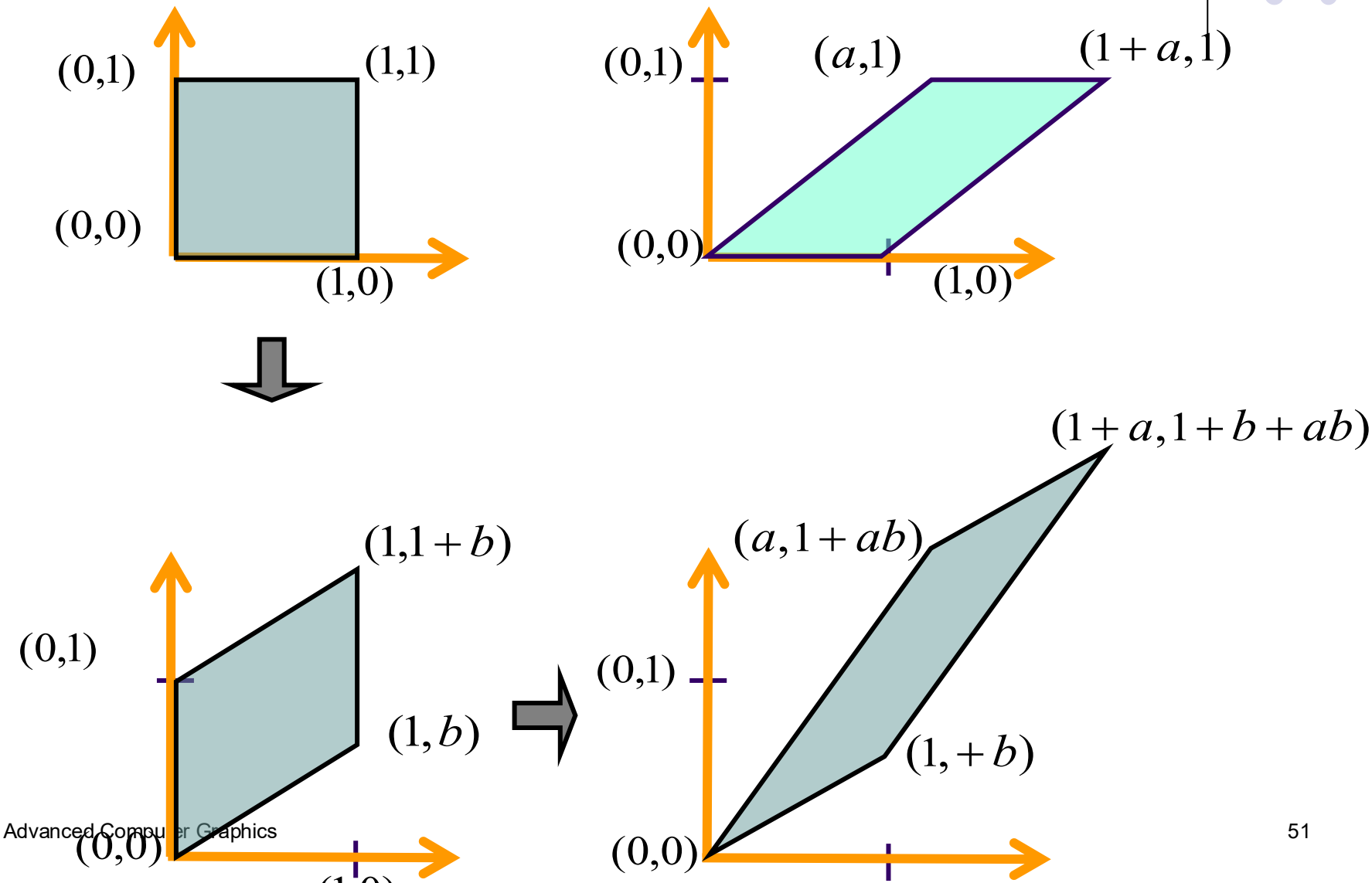


$$Sh\ in\ y_{(b=1)}\ ,\ \text{then}$$
$$Sh\ in\ x_{(a=1/2)}$$

Shear in x then *in y*

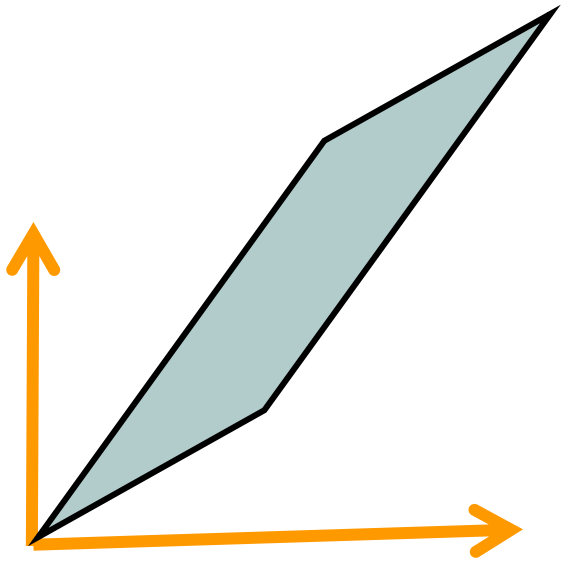


Shear in y then *in x*

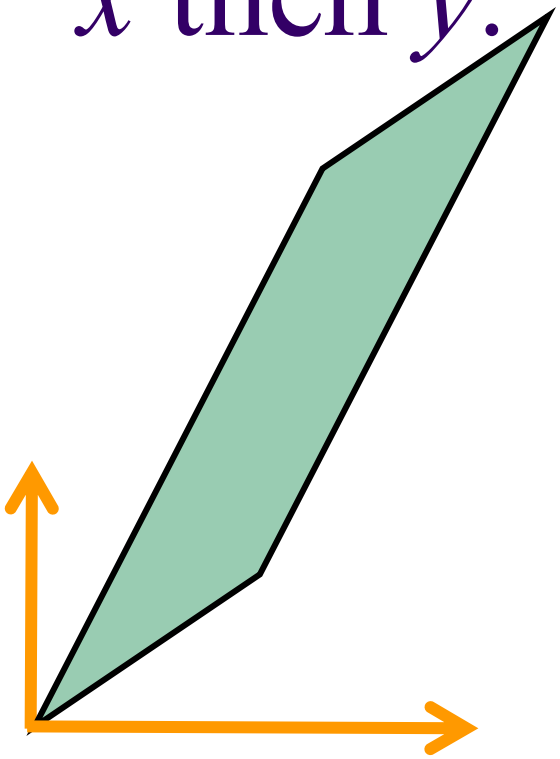


Results Are Different

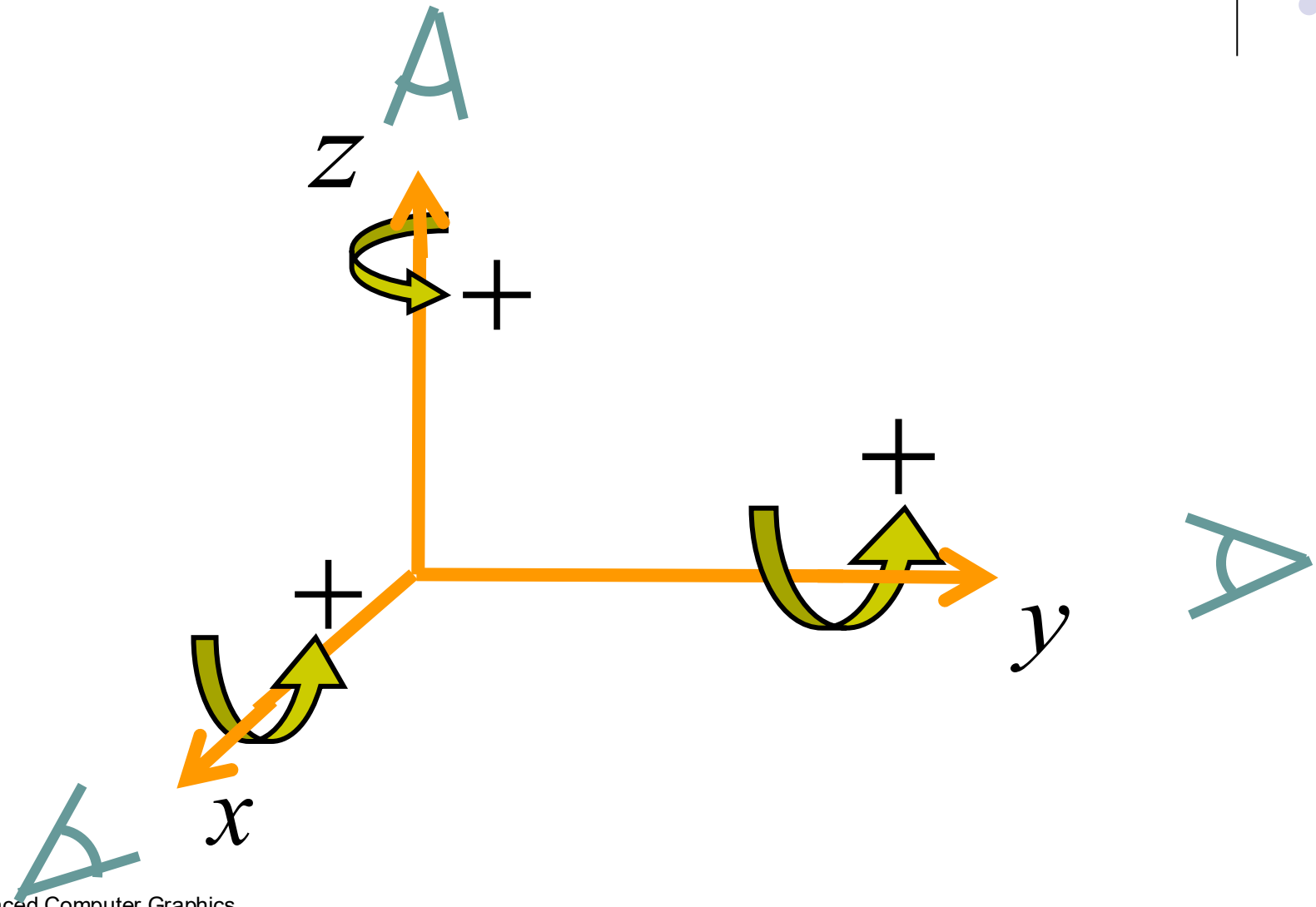
y then x:



x then y:



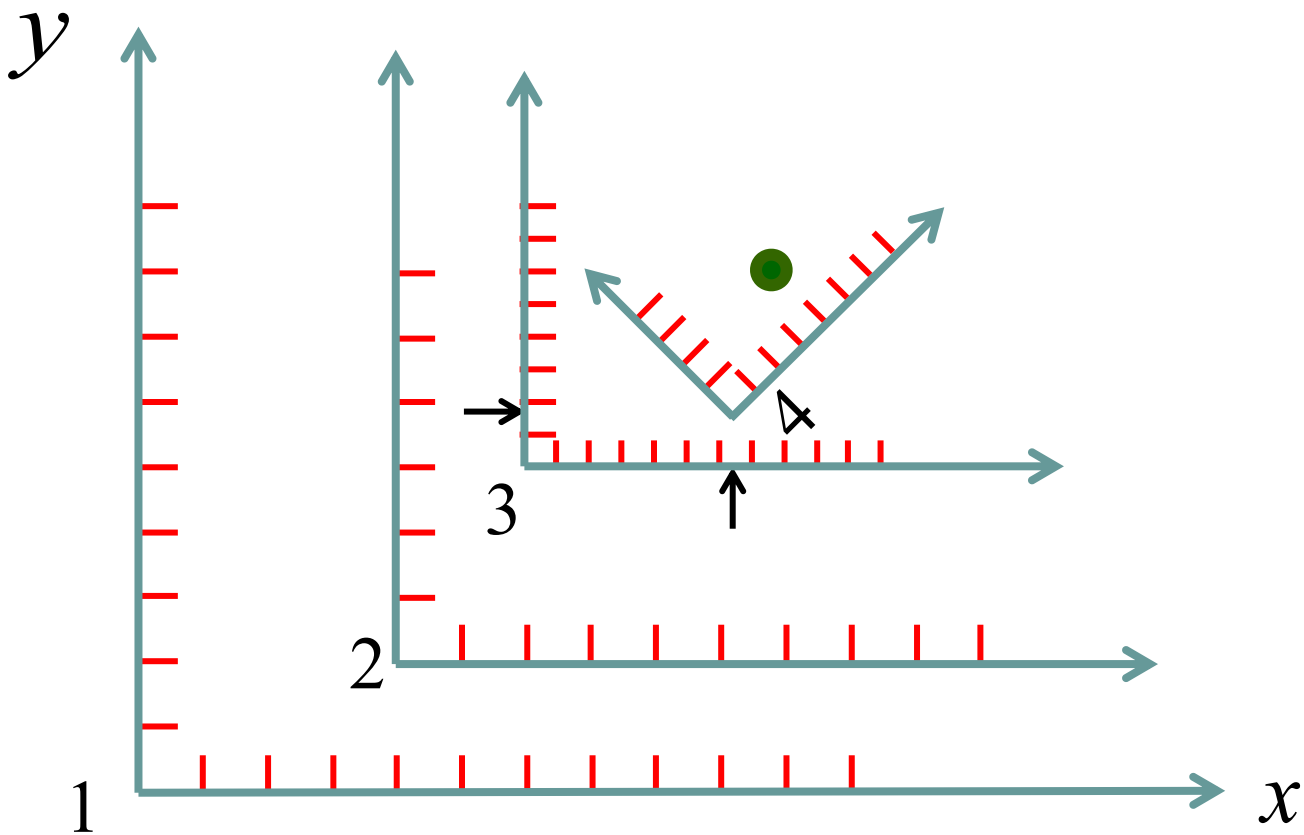
3D Positive *Rotations*



Transf's as Change in Coordinate Sys

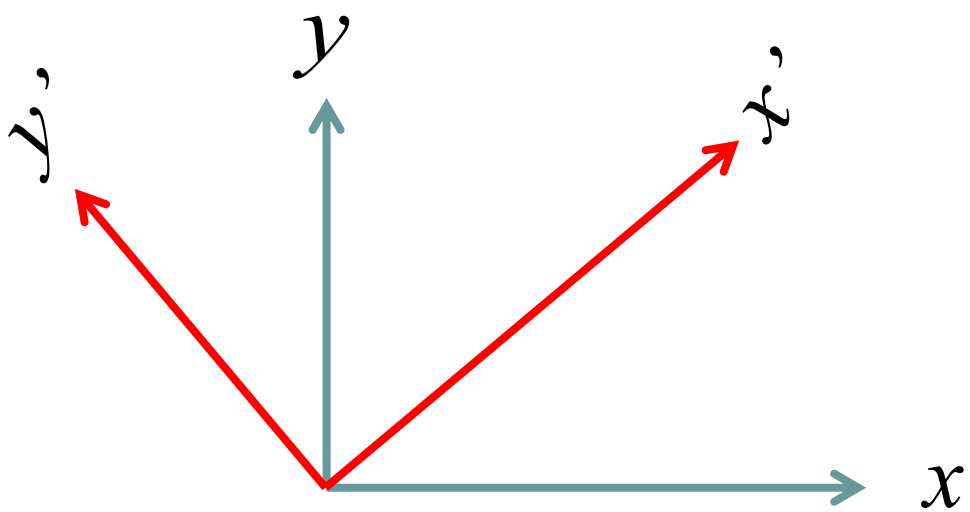
- Useful in many situations
- Use most natural coordination system locally
- Tie things together in a global system

Example



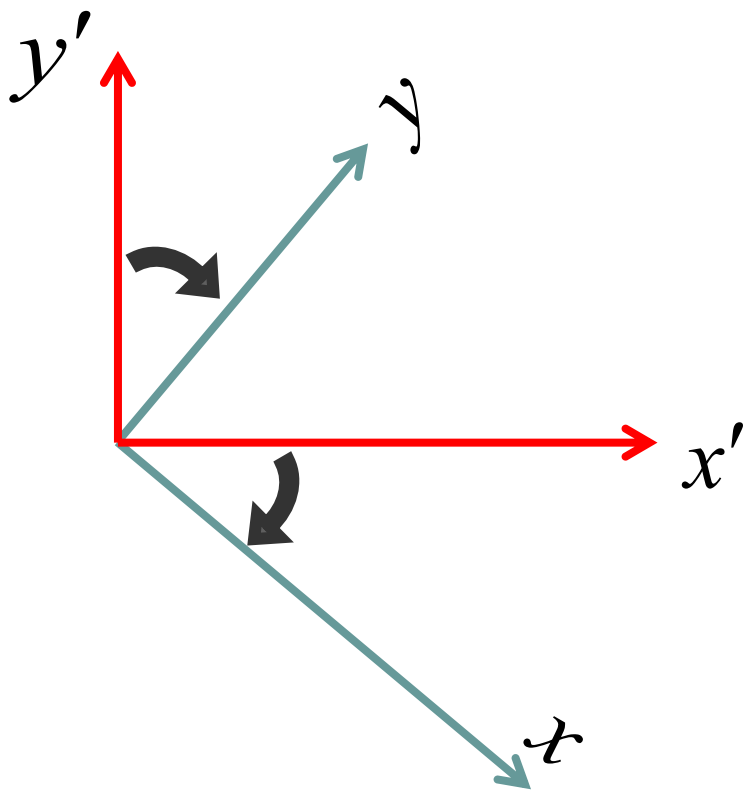
Change of Coordinate System

- Describe the *old* coordinate system in terms of the *new* one.



Change of Coordinate System

Move to the *new* coordinate system and describe the one *old*.



Old is a negative rotation of the *new*.

What is "*Perspective*?"

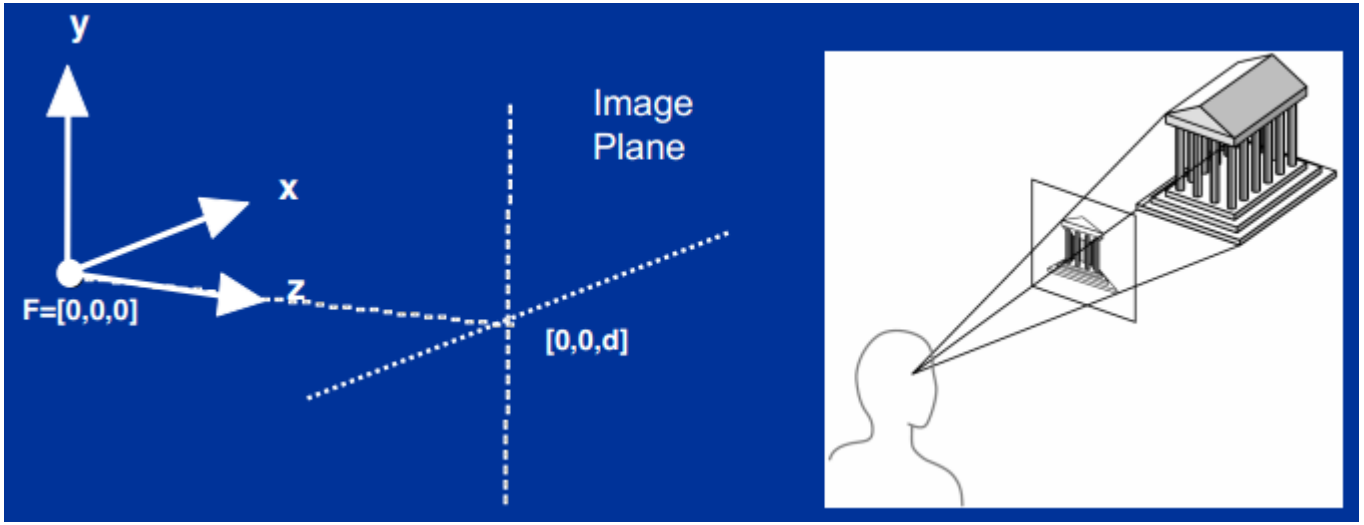
- A mechanism for portraying $3D$ in $2D$
- "*True Perspective*" corresponds to projection onto a plane
- "*True Perspective*" corresponds to an ideal camera image
- ..more detail: http://en.wikipedia.org/wiki/Perspective_(graphical)

Perspective Projection

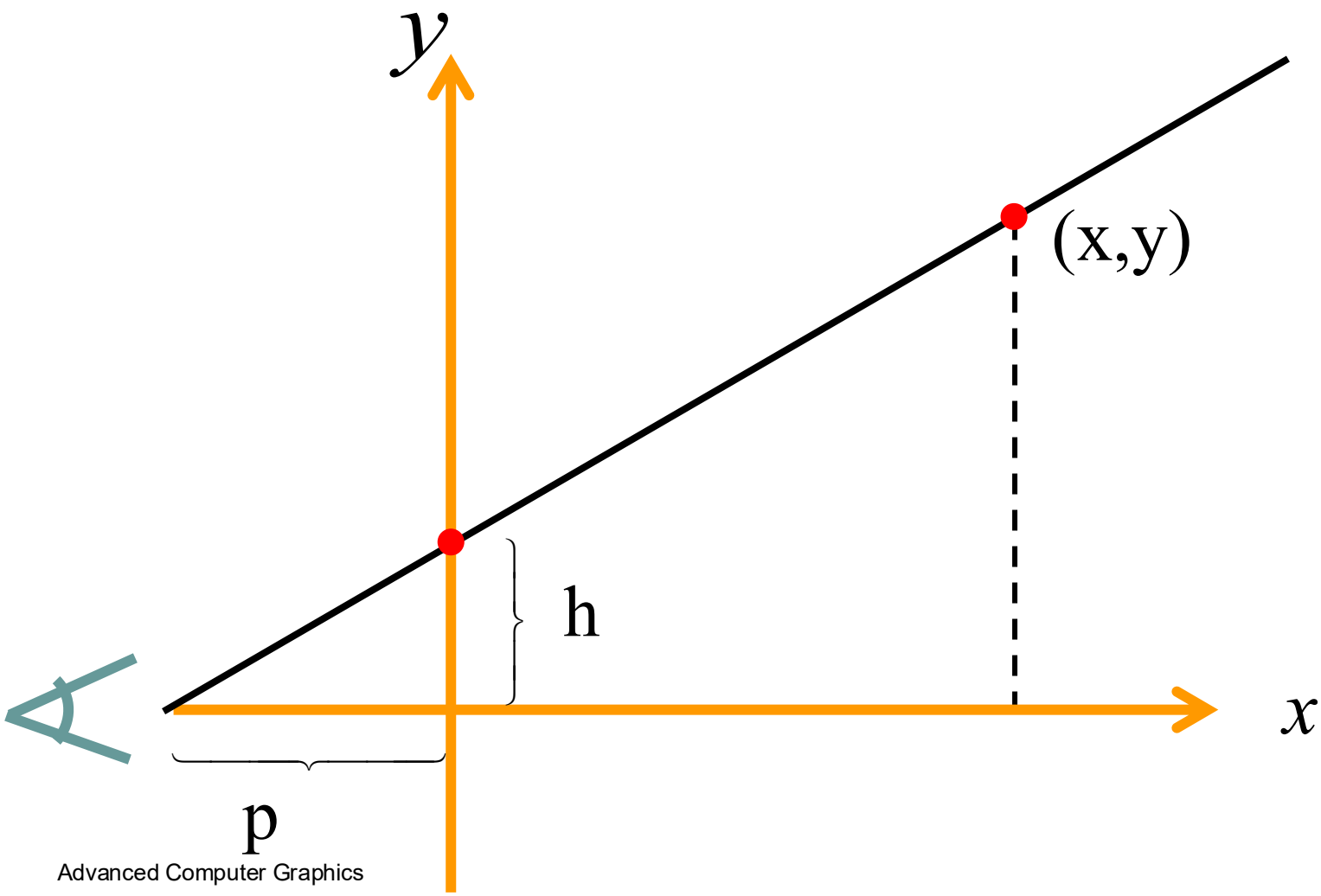
- With a perspective projection, objects become smaller as they get further from the camera, as one normally gets from a photograph or video image
- The goal of most real-world camera lens makers is to achieve a perfect perspective projection
- Most realistic computer generated images use perspective projections
- A fundamental property of perspective projections is that parallel lines will *not* necessarily remain parallel after the transformation (making them non-affine)

Simple Perspective Camera

- Canonical case:
  - camera looks along the z-axis
  - focal point is the origin
  - image plane is parallel to the xy-plane at distance d (We call d the focal length, mainly for historical reasons)



“True” Perspective in 2D



“True” Perspective in $2D$

$$\frac{h}{p} = \frac{y}{x+p}$$

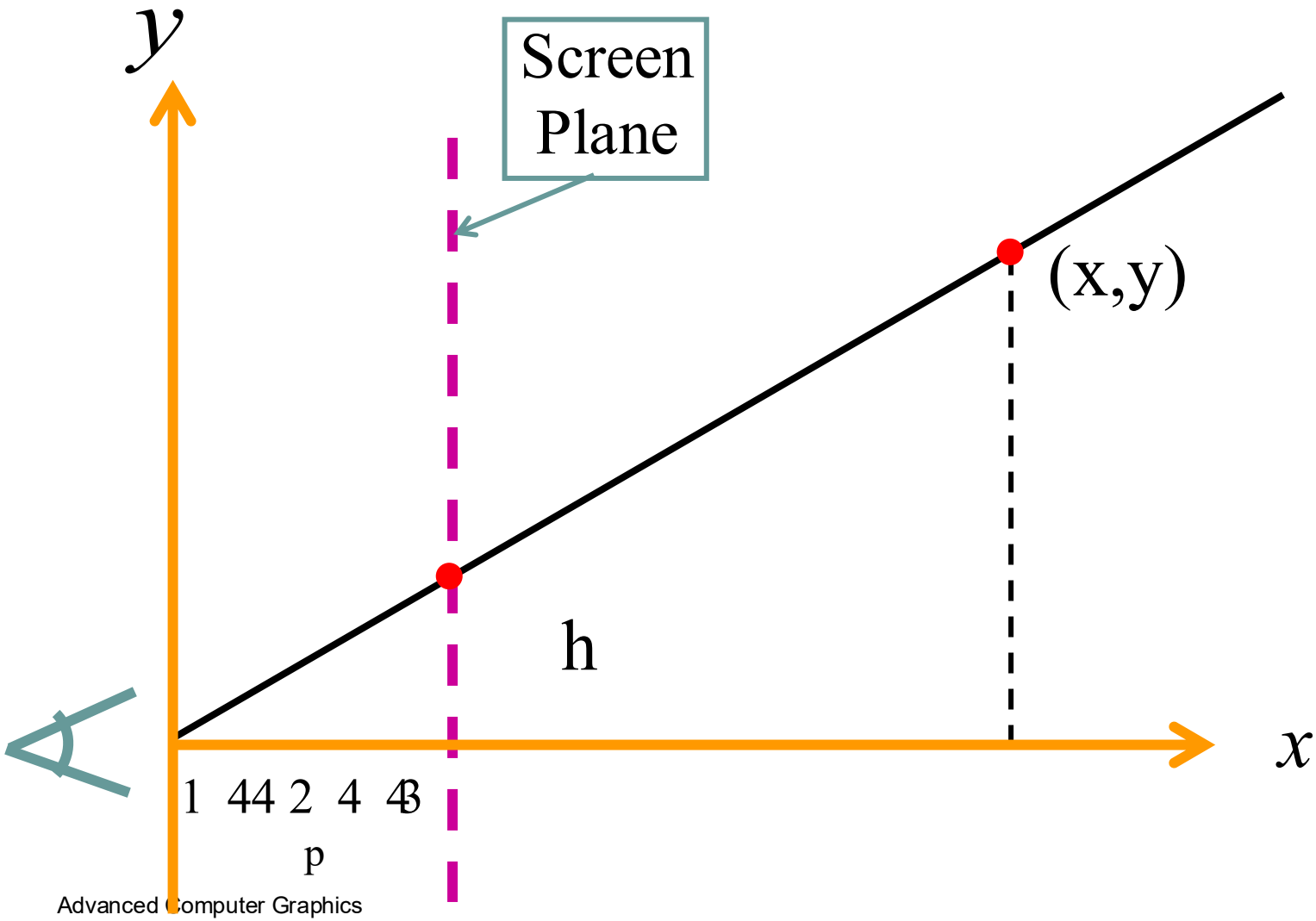
$$h = \frac{py}{x+p}$$

“True” Perspective in $2D$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1/p & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x \\ y \\ x/p + 1 \end{bmatrix}$$

$$= \begin{bmatrix} x \\ y \\ \frac{x+p}{p} \end{bmatrix} = \begin{bmatrix} \frac{px}{x+p} \\ \frac{py}{x+p} \\ 1 \end{bmatrix} \Leftrightarrow \begin{bmatrix} \frac{px}{x+p} \\ \frac{py}{x+p} \end{bmatrix}$$

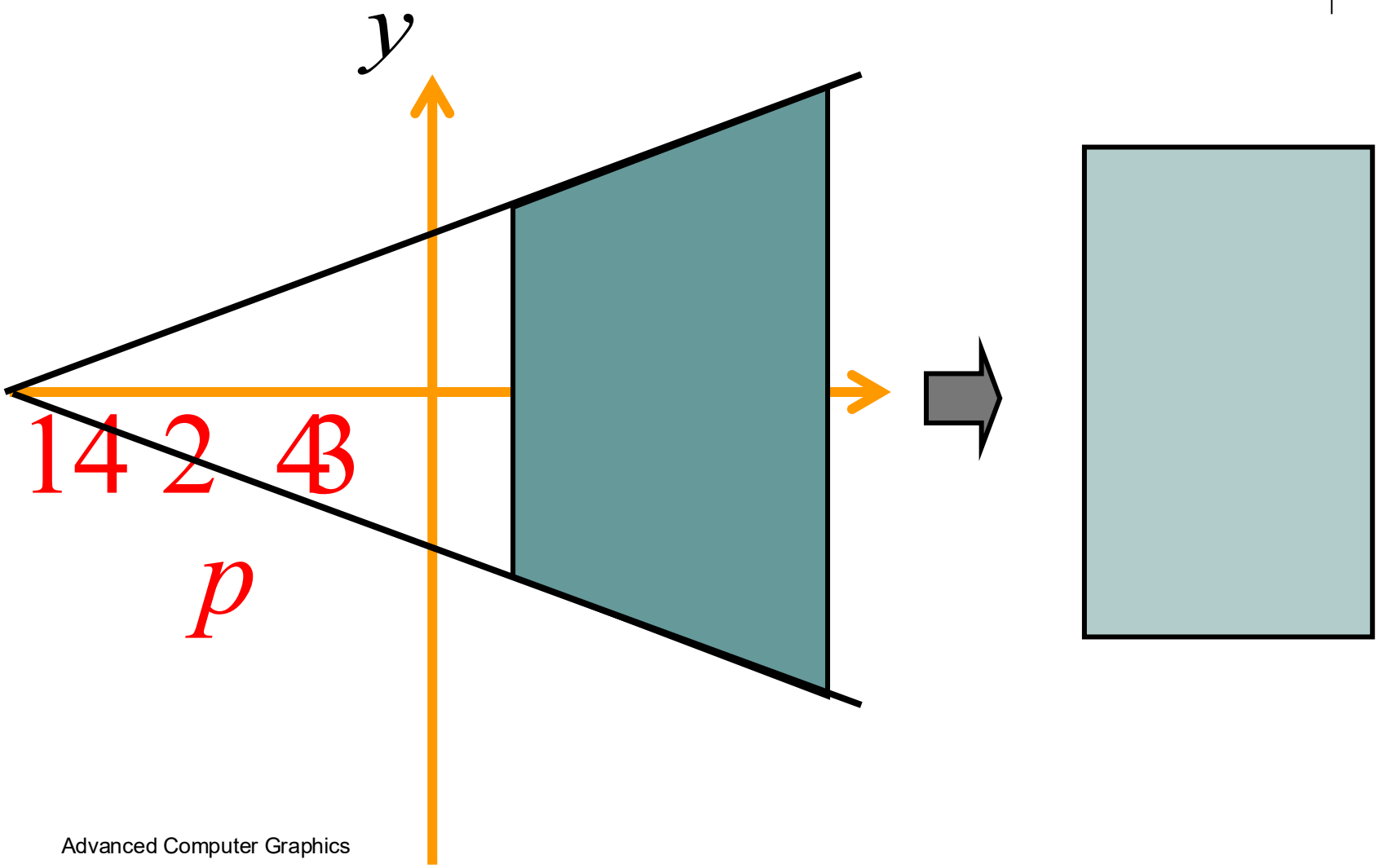
Geometry Same for Eye at Origin



Let's Look at a Limit

Observe,

$$\lim_{n \to \infty} \begin{bmatrix} 1 \\ 0 \\ \left(\frac{1}{n}\right) \end{bmatrix} = \begin{bmatrix} n \\ 0 \\ 1 \end{bmatrix} \Leftrightarrow \begin{bmatrix} n \\ 0 \end{bmatrix}$$

We see that $\begin{bmatrix} n \\ 0 \end{bmatrix} \Leftrightarrow +\infty$ on x-axis

“True” Perspective in *2D*